Problem set 4: (Addiitional problems, pick one)

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Gauti Eggertsson

Problem 1

Solve problem 1 in Chapter 10

Problem 2. Solving a linear rational expectation model using method of Blanchard and Kahn

Consider a log-linear model consisting of the following equations:

$$\hat{m}_t = \hat{Y}_t$$

$$\pi_t = \kappa \hat{Y}_t + \beta E_t \pi_{t+1}$$

$$\hat{\mu}_t = \rho \hat{\mu}_{t-1} + \epsilon_t$$

Here m_t referes to real money balances [$= M_t/p_t$, where M_t is the money supply and p_t is the price level],Y_t to the level of output, π_t to the gross rate of inflation [$= p_t/p_{t-1}$], and μ_t to the gross rate of growth of money supply [M_t/M_{t-1}]. In the case of each of these variables, a hat refers to the deviation of the value of the log of the variable from the value of the log of the variable in steady state that would exist in the absence of shocks to the economy. [Thus $\hat{m}_t$ refers to $logM_t - \log p_t - \log m^*$, where m^* is the steady-state level of real balances, for example.] In these equations, the coefficients are assumed to satisfy $\kappa > 0$, $0 < \beta, \rho < 1$ and $\{\epsilon_t\}$ is a sequence of i.i.d. random variables, with bounded support and mean zero, representing monetary policy disturbances. Equatinos (1)-(3) must be satisified at each data $t \geq 0$, with the values $\hat{m}_{-1}, \hat{\mu}_{-1}$ given as initial conditions.

[Equation (1) is a money demand equation that assumes a constant velocity of money; it also amounts to a monetaris specificiation of aggregate demand in this economy. Equation (2) is an aggregate supply relationship of the kind we have developed in class, derived from firms pricing decisions. Finally equation (3) specifies monetary policy, in terms of an exogenous stochastic process for the growth of money supply. Note that the assumption stated above imply that the rate of growth of the money supply μ_t is a stationary random variable with a bounded support, and that the money supply M_t will be poisitive at all times.]

(a) Show that there exists an equilibrium in which the variables $\hat{m}_t, \hat{Y}_t$, and π_t are each a linear function of $\hat{m}_{t-1}$ and $\hat{\mu}_t$, and in which each of these variables is stationary. Describe the equilibrium by specifying the two coefficients in each of the linear functions. [Hint: Note that the variables m_t, μ_t, and π_t are linked by an identity.]

If you can, show also that this solution represents the unique equilbrium in which each of these variables is bounded [which would be necessary for a solution to the approximate condition (2) to approximate a solution to the exact equilibrium condition].

(b) Compute the impulse response functions indicating the response of real output, the price level, and the nominal interest rate to a monetary shock ϵ_t, in all periods $T \geq t$. [Let $\epsilon_t = 1$]. For which values values of the model parametes, if any, does a monetary shock result in a "hump-shaped" output response of the kind indicated by the VAR estimates of Christiano et al.?

Problem 3: Deriving the New Classical Phillips curve via linear approximation

Time is discrete. There is a continuum of firms of measure 1 each one producing a good of variety j and facing the demand function

$$y_t(j) = (\frac{p_t(j)}{P_t})^{-\theta} Y_t$$

where $y_t(j)$ is the demand of firms j production, $p_t(j)$ its price, $\theta > 1, Y_t$ the Dixit-Stiglitz aggregate defined by

$$Y_t \equiv (\int_0^1 y_t(z)^{\frac{\theta-1}{\theta}} dz)^{\frac{\theta}{\theta-1}} \tag{1}$$

and P_t the Dixit-Stiglitz price index defined by

$$P_t \equiv (\int_0^1 p_t(z)^{1-\theta} dz)^{\frac{1}{1-\theta}} \tag{2}$$

We assume that firms have the production function

$$y_t(j) = A_t h_t(j)^{\alpha}$$

with $0 < \alpha < 1$

A fraction γ of firms set their prices one period in advance (sticky prices), a fraction $1-\gamma$ set their prices freely in every period (flexible prices) i.e. after date t market conditions are known, taking aggregate output and the aggregate price level as given. Each firm hires labor $h_t(j)$, facing common nominal wage rate W_t that it takes as given. Firms maximize profits, **using** the nominal stochastic discount factor

$$Q_{t,t+1} = \beta \frac{Y_t}{Y_{t+1}} \frac{P_t}{P_{t+1}}$$

In equilibrium real wages are given by

$$\frac{W_t}{P_t} = \chi Y_t^{1+\omega} A_t^{-\omega}$$

where $\omega, \chi > 0$. Aggregate nominal spending (demand) is exogenous and given by

$$D_t \equiv Y_t P_t \tag{3}$$

a. Write down the first order condition of the optimal choice of price $p_t(j)$ for a given firm j (i) under the assumption that prices are set at time t (ii) under the assumption that the prices are set at time $t-1$. Use the assumptions above in order to express these first order conditions as relationship between $p_t(j), P_t$, Y_t and A_t.[They should be of the form

$$f(p_t(j), P_t, Y_{t,} A_t) = 0 \tag{4}$$

$$E_{t-1} g(p_t(j), P_t, Y_{t,} A_t) = 0 \tag{5}$$

Hint: Under this pricing assumption you only need to consider the period profits for flexible prices, and one period in advance profits for the sticky prices]

b. Let p_t^s be the price charged in period t for the sticky-price good and p_t^f be the price charged for all flexible price goods. Using the result from part (a): (i) write a complete system of equilibrium conditions to determine p_t^s, p_t^f, P_t, and Y_t, given the exogenous process for productivity, A_t, and D_t.[You may state the equilibrium in form of equations (4) and (5) in case you did not complete (a)] and (ii) define the *natural rate of output.*

c. Do a first order approximation of the system around a steady state in which $\frac{P_t}{P_{t-1}} = 1$, $Y_t = \bar{Y}$ in (b) to yield

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) + E_{t-1}\pi_t \tag{6}$$

from the pricing equations for the firms where $\hat{Y}_t \equiv \frac{Y_t - \bar{Y}}{\bar{Y}}$ denotes output in percentage deviation from steady state, $\hat{Y}_t^n$ the natural rate of output and π_t inflation. What is the epression for κ an Y_t^n? Suppose that nominal demand follows the exogenous process

$$\hat{D}_t = \rho \hat{D}_{t-1} + \epsilon_t \tag{7}$$

where $\hat{D}_t$ is nominal demand, defined in (3), in percentage deviation from steady state. Linearize equation 3 and use the resulting equation, together with (6) and (7) to define an approximate equilibrium for $\{\pi_t, \hat{Y}_t, \hat{D}_t\}$.(i) How does a shock to nominal demand affect $E_t \hat{Y}_{t+j}$? (ii) Without explicitly deriving the relevant expression (although you may do so) on what shocks does the natural rate of output depend upon? iii) How does the parameter κ depend on the number of flexible prices?

Problem 4: Optimal policy under discretion and commitment.

Consider the following approximation of a New Keynesian rational expectations model that we studied in class.

$$U_t = -\sum_{t=0}^{\infty} \beta^t \{\pi_t^2 + \lambda_x {x_t}^2 + \lambda_i \hat{i}_t^2\}$$

$$x_t = E_t x_{t+1} - \sigma(\hat{\imath}_t - E_t \pi_{t+1} - \hat{r}_t^n) \tag{8}$$

$$\pi_t = \kappa x_t + \beta E_t \pi_{t+1} \tag{9}$$

where $\hat{r}_t^n$ is an exogenous process that follows

$$\hat{r}_t^n = \rho \hat{r}_{t-1}^n + \epsilon_t$$

and ϵ_t is iid. Here π_t is inflation, x_t is the output gap, $\hat{\imath}_t$ is deviation of the nominal interest rate from steady state, and $\lambda_x, \lambda_i, \kappa, \sigma > 0$, $0 < \beta, \rho < 1$ are coefficients. U_t is the objective function the government want to maximize, part of which corresponds to a second order expansion of the household utility. The only difference between this model and the one we studied in class is the presence of the *ad hoc* interest rate stabilization term in the objective function of the government (although a similar term can be motivated from microfoundations if one adds a role for transaction frictions in the model. A microfounded term, however, would depend on the absalute devition of the nominal interest rate from zero, rather than its deviation from steady state).

(a) Write the first order conditions for the government's problem from period $t \geq t_0$ onward if it can commit to future policy. Define the equilibrium in terms of sequences of the endogenous variables, given the sequence for the exogeneous disturbances. Discuss informally (or formally if you wish) the restriction on the optimal plan imposed by the "time-less perspective". [Hint: Note that in class we could ignore equation (8) in writing the government's problem. This is not longer case once we include $\hat{\imath}_t$ in the objective function.]

(b) Write down the first order conditions for the government's problem in a Markov Perfect Equilibrium, i.e. if it cannot commit to future policy. This is optimal policy under "discretion". [Hint: In this case the expectation's can be written as functions of the state, e.g. $E_t \pi_{t+1} = \bar{\pi}(r_t^n)$, that the government takes as exogenous].

(c) OPTIONAL, you may need to use a software for this – Plot up the impulse response function of π_t, x_t and $\hat{\imath}_t$ to an innovation in $\hat{r}_t^n$. To do this you can assume the following parameters. $\lambda_x = \lambda_i = 1$. $\sigma = 1$, $\kappa = 0.02$, $\beta = 0.99$, $\rho = 0$. How does discretion differ from commitment? Is there a dynamic inconsistency problem in this policy problem? If there is difference, what is the intuition?

Problem 5

A representative household maximizes expected utility according to the utility function

$$E_t \left\{ \sum_{T=t}^{\infty} \beta^{T-t} b_T [\log C_T - \psi \frac{h_T^{1+\omega}}{1+\omega}] \right\} \tag{10}$$

where b_t is an intertemporal shock, C_t is a Dixit-Stiglitz aggregate of consumption of each of a continuum of differentiated goods

$$C_t \equiv [\int_0^1 c_t(i)^{\frac{\theta_t}{\theta_t - 1}}]^{\frac{\theta_t - 1}{\theta_t}}$$

with the time-varying elasticity of substitution equal to $\theta_t > 1$, P_t is the Dixit-Stiglitz price index,

$$P_t \equiv [\int_0^1 p_t(i)^{1-\theta_t}]^{\frac{1}{1-\theta_t}}$$

and h_t is hours worked. E_t denotes mathematical expectation conditional on information available in period t.

For simplicity only one period riskless government bonds are traded. The household budget constraint can be written in term of the consumption aggregate as

$$P_t C_t + B_t = (1 + i_{t-1})B_{t-1} + Z_t + P_t w_t h_t - P_t T_t \tag{11}$$

where Z_t is a representative firm profit, T_t taxes, B_t one period riskless bonds, i_t one period nominal risk-free interest rate and w_t real wages.

The Dixit-Stiglitz market structure implies a demand for each good variety of $c_t(i) = (\frac{p_t(i)}{P_t})^{-\theta} C_t$ There is a continuum of firms on the unit interval that maximize expected discounted profits, each producing one of the good varieties. Firms produce using a production function

$$y_t(i) = h_t(i)$$

As Rotemberg (1983), we assume that firms face a resource cost of price changes $\frac{\delta}{2}(\frac{p_t(i)}{p_{t-1}(i)} - 1)^2$. The firms hire labor from the households, taking wages, w_t, as given.

a). Write down the households maximization problem, i.e. derive the first order condition that determine the choice of the variables C_t,h_t, and B_t.

b) Write down the profit function of the representative firm.

c) Derive the first order condition for the firms maximization problem, assuming the firms discounts its profits using the marginal utility of the representative household, λ_t, which corresponds to the Lagrangian multiplier on 11 in the household maximization problem.

d) Aggregate the model by assuming a symmetric equilibrium, i.e. all firms charge the same price and note that all production is consumed.[Hint: In the end you should be able to write the consumption Euler equation and firms Euler on the following non-linear form

$$Y_t^{-1} = (1 + i_t) E_t Y_{t+1}^{-1} \Pi_{t+1}^{-1} \beta \frac{b_{t+1}}{b_t}$$

$$\Pi_t(\Pi_t - 1) = \frac{\theta_t}{\delta}(\psi Y_t^{1+\omega} - \frac{\theta_t - 1}{\theta_t})Y_t + Y_t E_t \Pi_{t+1}(\Pi_{t+1} - 1) Y_{t+1}^{-1} \beta \frac{b_{t+1}}{b_t}$$

where $\Pi_t = \frac{P_t}{P_{t-1}}$]

e) Now assume that the central bank can determine the nominal interest rate, i_t, and follows the policy rule

$$i_t = \phi(\Pi_t, b_t)$$

d) List set of equation that define an equilibrium and the sequnce of endogenous variables and exogenous one that define the equilibrium [you can use the equations from the hin tin d) if you had problem with the first part. The endogenous ones are $\{i_t, Y_t, \Pi_t\}$ while the exogenous ones are $\{b_t, \theta_t\}$]. Define the steady state that is consistent with zero inflation, i.e. $\Pi_t = 1$

e) Linearize the model around this steady state. Define each variable in terms of percentages deviation from steady state. In particular,

$\pi_t = \frac{\Pi_t - 1}{1}$, $\hat{Y}_t = \frac{Y_t - \bar{Y}}{\bar{Y}}$, $\hat{\imath}_t = \frac{i_t - \bar{\imath}}{1+\bar{\imath}}$. Define the approximated equilibrium for $\{\hat{\imath}_t, \hat{Y}_t, \pi_t\}$ given the exogenous shocks.

2. You should be able to express you model in the following form

$$\hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma(\hat{\imath}_t - E_t \pi_{t+1} - \hat{r}^e_t)$$

$$\pi_t = \kappa \hat{Y}_t + \beta E_t \pi_{t+1} + \omega_t$$

$$\hat{\imath}_t = \hat{\gamma}_t + \phi_\pi \pi_t$$

a) what is the value of each of the parameters above, σ, κ? What is the interpretation of the shocks r^e_t and ω_t. How does the value of δ relate to the frequency of changing prices in the Calvo model?

b) We now want to do a simple exercise related to the zero bound. To simplify notation, subtract the constants from $\hat{\imath}_t$, $\hat{r}^e_t$ and $\hat{\gamma}_t$ so that they now are written in terms of their absolute value i_t, r^e_t and γ_t. For simplicity suppose that $\gamma_t = \beta^{-1} - 1 = \bar{r}$ The model is then written as

$$\hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma(i_t - E_t \pi_{t+1} - r^e_t)$$

$$\pi_t = \kappa \hat{Y}_t + \beta E_t \pi_{t+1} + \omega_t$$

$$i_t = \max(0, \bar{r} + \phi_\pi \pi_t)$$

Consider now the simple stochastic process for r^e_t that it is $r^e_L < 0$ in period 0 and revert back to steady state $\bar{r}$ with fixed probability α in every period. Once it goes back to steady state it stays there forever. Let us call the data the shock reverts back to steady state τ.

Consider first the solution at date $t \geq \tau$. Assume that ω_t is iid with mean zero in $t \geq \tau$.. *Assuming* the zero bound will no longer be binding at this stage, for what value of ϕ_π does there exists a unique bounded solution at positive interest rate for output and inflation? What is this solution? How does output

depend upon ω_t? What is the intuition? Does an increase in monopoly power increase or decrease equilibrium output?

c) Now consider the solution for $t \leq \tau$. Let us assume that during this period the shock ω_t also takes on a value $\omega_L > 0$ that is perfectly correlated with r_L^e (i.e. it reverts back to being a mean zero iid process with probability α every period like the r_L^e reverts to steady state of $\bar{r}$ with that same probability).

Suppose that r_L^e and ω_L are such that the zero bound is binding in period $t < \tau$. What does the solution in period $t < \tau$ look like and what bounds on α are needed to ensure a unique solution?. How does the solution depend on ω_L? Does an increase in monopoly power increase or reduce output at the zero bound?

Problem 6

Solve problem 1 in Chapter 10